

STAT1008 Quantitative Research Methods
Group Presentation Project

OBJECTIVES

The main goal of the group presentation project is to help you apply the statistical tools that you have learned in this course, and to do so in a realistic data analysis setting. The project also requires you to work collaboratively in groups to analyse data and orally present your findings (see Learning Objective 2).

GROUP FORMATION REQUIREMENTS

- **By Friday of Week 3 (14th August), 5pm:** Form a team of four students – all members must be registered in the same computer lab – email your tutor once your team is formed.
- If your team has fewer than three members or more than four, or if you don't email your tutor by the deadline, you will be randomly assigned additional team members or placed into a new team to ensure all groups have three to four members.
- This assessment cannot be done individually.

ORAL PRESENTATION GUIDELINES AND PRESENTATION ORDER

- **Expected Presentation Length:** 12-15 minutes. Your tutor will set a 15-minute alarm at the start of your presentation. When the alarm sounds, you must stop immediately. Plan accordingly. Aim for 1 to 1.5 minutes per slide.
- All team members must present and participate approximately equally.
- Absent members or members who do not speak receive zero marks.
- The tutor may ask one or two questions at the end of the presentation. There will be no time for questions from the audience.
- Tutors will randomly allocate each group to a presentation week (either Week 9, 10, 11, 12). Groups will present in their computer lab time of their allocated week. Individual requests for a preferred presentation week will not be considered for the random allocation.
- Presentation orders will be published by Week 5
- If you have a conflict and can find a willing team within your computer lab to swap with you, both teams can approach the tutor to register a swap. This must be done by two weeks before the earlier of the two originally scheduled dates. No late swapping is permitted.

PROJECT REQUIREMENTS

Each group is required to analyse a dataset of their choice using the statistical tools and methods discussed in this course. The analysis must include some exploratory data analysis and one hypothesis test. Each group will present their findings in an oral, in-person presentation during their computer lab time in either Weeks 9, 10, 11 or 12. The presentation time will be randomly allocated by the tutor (see page 1).

Each group will need to decide on what data summaries and graphical displays to produce, and what numerical descriptive measures to calculate for inclusion in the presentation. For the hypothesis test, each group will need to decide on what applied question to answer with the chosen dataset, formulate the question in terms of a hypothesis test, provide step-by-step details of the calculations, and interpret the analysis results in relation to the original question of interest. Specific requirements of the presentation are described in detail on page 4.

Listed below are a few example applied questions that could be investigated:

- *Are Top 100 songs shorter than average compared to the Top 100 songs twenty years ago?* (This question is related to the shortened attention span of the current audience <https://www.bbc.co.uk/news/resources/idt-052ab668-403d-416f-b5a6-c5692313b9b4>).
- *Is there a significant relationship between pet ownership rates and fertility rates by country?* (As fertility rates drop around the world, are young people spending vast amounts of money on raising pets instead of children?)
<https://www.dw.com/en/japan-are-pets-replacing-children-during-the-pandemic/a-57669883>).

This is a group assessment, so it is very important to try and select a research question and find relevant data that all group members are excited to work with!! The group project will not be successful if any member is unhappy with the dataset choice, so please collaborate, participate and work respectfully with each other. The dataset can come from any field such as economics, law, medicine, education, sport, psychology, politics etc. Your analysis must be original and must not be copied from another source. Once you have chosen your dataset, you may wish to confirm with your tutor that your choice of dataset is suitable in terms of complexity and structure.

CAUTIONS!!

- YOU MUST ANALYSE RAW, INDIVIDUAL LEVEL DATA (NOT AGGREGATED DATA). DO NOT ANALYSE DATA ALREADY SUMMARISED IN A FREQUENCY OR SUMMARY TABLE. (Note: data tables available for public download from the Australian Bureau of Statistics tend to be in aggregated form already, so this data is not suitable for the project. An example of a data table that is already aggregated is shown in the screenshot on the following page).
- THE STATISTICAL HYPOTHESIS TESTING TOOLS WE LEARN IN THIS COURSE REQUIRE INDEPENDENT UNITS OF OBSERVATIONS. FOR EXAMPLE, SERIALY CORRELATED DATA (that is, data taken at yearly or other regular time intervals) ARE NOT INDEPENDENT. So time series data (eg. share prices over time) are not suitable for the assignment.

Australian Bureau of Statistics Australian Bureau of Statistics						
9208DO004_202006 Electric passenger vehicle use, experimental estimates, 12 months ended 30 June 2020						
Released at 11:30 am (Canberra time) 26 November 2021						
Table 1 Summary of passenger vehicle use, by state/territory of registration by type of fuel						
	Total kilometres travelled	Total kilometres travelled - RSE	Number of vehicles	Number of vehicles - RSE	Average kilometres travelled	Average kilometres travelled - RSE
	million	%	no.	%	'000	%
New South Wales						
Petrol	39,282.1	7.44	3,751,354	3.11	10.5	6.70
Diesel	7,003.3	21.34	592,754	16.97	11.8	14.58
Electric	23.9	16.95	2,094	12.22	11.4	12.81
LPG/CNG/dual fuel/hybrid and other	605.8	69.47	25,131	93.13	24.1	84.02
Total	46,915.1	6.23	4,371,333	1.51	10.7	6.13
Victoria						
Petrol	35,786.5	7.00	3,380,124	3.59	10.6	6.09
Diesel	7,259.8	25.16	473,099	19.17	15.4	16.00
Electric	25.2	19.41	2,256	16.23	11.2	13.18
LPG/CNG/dual fuel/hybrid and other	2,302.5	41.02	124,394	37.80	18.5	28.43
Total	45,373.9	5.65	3,979,873	1.47	11.4	5.46
Queensland						
Petrol	25,508.5	6.54	2,430,821	2.98	10.5	5.94
Diesel	6,183.9	19.94	446,599	14.54	13.9	13.39
Electric	10.0	7.33	987	7.28	10.1	6.67
LPG/CNG/dual fuel/hybrid and other	860.0	41.05	50,903	42.02	16.9	37.23
Total	32,562.2	5.40	2,929,310	1.58	11.1	5.32

Figure 1: Example of aggregated (not individual level) data

DATA SOURCES

If you have a specific topic or data field in mind, try running an online search for your topic eg. `basketball data`, `human rights data`, `cost of living data` .

For a general search of datasets from a variety of topics, the following websites may be useful:

- UC Irvine Machine Learning Repository <https://archive.ics.uci.edu/ml/index.php>
- The Data and Story Library <https://das1.datadescription.com/>

For those interested in global and country-level data, you may find the following websites useful:

- Our World in Data <https://ourworldindata.org/>
- OECD Data <https://data.oecd.org/>

As an alternative to sourcing datasets from the internet, you can collect your own data (by conducting your own survey for example). If collecting data from people, please respect the privacy and confidentiality of survey respondents. If you have concerns about your data collection method, please speak to your tutor. A minimum sample size of 30 is recommended for valid statistical inference. Please collect data on at least four variables for analysis.

PRESENTATION SLIDES - CONTENT

Each group must submit and orally present in-person a set of presentation slides to communicate their project findings. Please include the following sections in the presentation:

- **TITLE SLIDE:** Title of the presentation; List group members.
- **INTRODUCTION:**
 - State the research objective. What applied question are you trying to answer?
 - Explain why the research question is of personal and/or practical interest.
- **DATA SET DESCRIPTION:**
 - Dataset source.
 - * **Online sourced dataset:** Provide website address of source. What time period are the data applicable to? (eg GDP by country as at 31 Dec 2020). Discuss any potential survey errors; OR
 - * **Self-collected data:** What is the sampling design? How were people recruited for the survey (if applicable)? Discuss any potential survey errors.
 - Dataset size and variables. State the sample size. List and provide definitions of the variables analysed. Classify the variables by type (numerical, categorical etc....) Note: You do not need to analyse all the variables in your chosen dataset.
- **EXPLORATORY DATA ANALYSIS**
 - Provide summary (frequency or contingency) tables
 - Provide some graphical displays
 - Provide some numerical descriptive measures (eg, sample means, sample variances, sample proportions). You do not need to show working for any numerical descriptive measures that you calculate in this section. You can simply report the result.
 - From your data summaries, what conclusions can you draw about the shape of the distribution of your variables or relationships between variables? Try to explain any patterns in the data you notice.
- **HYPOTHESIS TEST :**
 - State H_0 : and H_A :
 - State the significance level.
 - Show and explain the test statistic calculation.
 - State the critical value(s) of the test.
 - State the p-value of the test and the mathematical expression to calculate it. Report the exact p-value calculated using the Excel function T.DIST(..) or NORM.S.DIST(...) as demonstrated in lectures.
 - State the conclusion of the test in the context of the original question. Discuss whether the result makes intuitive or practical sense or not. For example, *a significant positive correlation is found between exercise hours and sleep hours. This makes sense as more rest time may be required to recover from the additional physical exertion during exercise.*

- **LIMITATIONS AND FUTURE WORK:**

- Discuss any limitations of your analysis and potential future improvements. Are there any further questions you would like to answer if you had the relevant statistical knowledge or if you had access to additional data?

- **REFERENCE LIST:** If applicable, please use the Harvard referencing style as detailed here <https://www.anu.edu.au/students/academic-skills/academic-integrity/referencing/harvard>.

SUBMISSION GUIDELINES ON CANVAS

- The presentation must be submitted online on the Canvas website via the Turnitin submission box by 5pm the day prior to your assigned presentation date. Failure to submit the presentation by this date will result in a penalty of 5% per day.
- You must also submit the Excel spreadsheet containing the raw data. **Failure to do so will result in a deduction of 2 marks.**

ACADEMIC INTEGRITY

- Groups should not have another person/entity do any portion of the project for them, which includes hiring a person or a company to complete any portion of the project.
- All parts of the presentation must uphold the principles of academic integrity, as defined in the ANU Policy: Code of Practice for Students University Academic Integrity Rule (<https://services.anu.edu.au/learning-teaching/academic-integrity>).
- Analytical and critical thinking skills, and effective communication of statistical ideas are part of the learning outcomes of this course. Developing strong competencies in these areas will prepare you for a competitive workplace. The presentation submitted must be original and the result of a group effort.
- All group members must participate in the preparation of the presentation. The presentation must not be produced from the efforts of one or two team members only.

USE OF ARTIFICIAL INTELLIGENCE (AI)

- It is acceptable to use AI tools (like ChatGPT) to (i) generate project ideas, and/or (ii) as an editing tool. If AI is used, you **must** include the following declaration after the 'Title' slide of the presentation:

We acknowledge the use of [*insert name of AI tool(s)*] to prepare this presentation.

AI was used to [*list the tasks you used AI for e.g to brainstorm ideas*].

- You should note that the material generated by AI programs may be inaccurate, incomplete, or otherwise problematic. Thus use of AI may result in a lower quality product with AI unable to produce the sophistication required for this task. AI is **not** a replacement for your own thinking and research.
- It is very important that you do not use AI to merely 'do' the entire project for you. Submissions that have been generated entirely by AI are not permitted and will be treated as plagiarism and a breach of ANU's Academic Integrity Rule.
- Failure to properly cite your use of AI as described above is also in violation of the ANU Academic Integrity Rule.

EXTENSION REQUESTS

If a team member encounters exceptional circumstances beyond their control such that they are unable to present on their assigned presentation date, this will affect the entire team. The team presentation can only go ahead if all team members are present. So team members must communicate with each other and decide whether or not to submit an Assessment Extension Request (using the link on the Canvas site).

To provide evidence that all group members have agreed to the assessment extension request, all group members must individually submit an extension request. In the request please provide the details of the team members, the assigned computer lab, and the team member who originated the team request. An Assessment Extension Request must be received by **5pm on the day prior to your assigned presentation date**. Assessment extension requests received after this time will be rejected. Assessment extension requests missing a submission by at least one team member will automatically be rejected. Submission of an extension request does not guarantee the extension will be approved. Extension requests will be reviewed by the course convenor in accordance with ANU Procedure: Student assessment (coursework) (https://policies.anu.edu.au/pp1/document/ANUP_004604).

If an extension request is approved by the course convenor, the entire team will be rescheduled to present, at a date and time and location set by the course convenor. This date and time will not be assigned or changed to fit the schedule of individual students or teams. Failure to attend the new assigned date will result in a grade of 0 for this assessment for all group members.

MARKING GUIDELINES (subject to minor changes)

The presentation will be marked according to the following rubric:

Item	Total marks available	Notes
Clarity	2	Are the presentation slides logically structured, easy to follow and neatly presented? Is the project title clear? Are graphs well labelled? Is the practical importance of the research objective clearly communicated? Did all members participate approximately equally in the presentation?
Dataset description	2	Were all the questions in this section addressed as required?
Data summaries	4	Were the relevant numerical descriptive measures/summary tables/graphs shown? Was valid commentary/interpretation provided on the data summaries and graphs?
Hypothesis test	3	Are your null and alternative hypotheses clearly stated using the statistical notation shown in lectures? Did you provide step-by-step calculation details including the mathematical expressions to calculate your test statistic and p-value? Did you calculate an exact p-value using the appropriate Excel function? Did you justify the assumptions required for the statistical methods you used?
Conclusions	2	Were valid limitations of the analysis discussed? Were valid potential future improvements discussed?
Individual presentation	2	Were the ideas communicated clearly by the speaker? Did the speaker establish eye contact with the audience? Did the speaker rely on notes or read verbatim from the slides?
Total	15	

NOTE: Your group must also submit the Excel spreadsheet with the raw data as a separate file. Failure to do so will result in a deduction of 2 marks. So if you score 15/15 on the presentation, but fail to submit your dataset, your final score will be 13/15.

Additional notes:

- Practise your presentation as a group beforehand. If possible, arrange to rehearse in front of another group, or video yourselves and review your presentation delivery. Confident and clear verbal delivery is just as important as the written content on your slides. If the material is not communicated clearly, the interpretation of your analysis may be misunderstood.
- Consult with the teaching staff for advice and to check whether your chosen dataset is appropriate or not.
- There is no need to use statistical methods that we haven't discussed in class yet.
- For copyright reasons and to avoid plagiarism, sample presentations will not be made available.
- The teaching staff will be happy to answer specific questions or concerns you have related to your project. But the teaching staff will not be available to review full drafts of your presentation before submission and provide detailed feedback. The teaching staff are not able to provide feedback on whether the depth of your analysis is sufficient for the project.